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Math 102 Final Exam Term 251

Original exam date: 22 December 2025. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Evaluate

$$\int_0^1 \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}}\, dx.$$

(a) $\frac{1}{2}\ln\left[2(e^2 + e^{-2})\right]$

(b) $\ln(e^2 + e^{-2})$

(c) $\frac{1}{2}\ln\frac{e^2 + e^{-2}}{2}$

(d) $\ln\left(\frac{e^2 + e^{-2}}{2}\right)$

(e) $\ln(e^2 - e^{-2})$

2. The area of the region bounded by the graphs of the equations $y = x^2 + 2x$ and $y = x + 2$ is

(a) $\frac{7}{2}$

(b) $\frac{11}{2}$

(c) $\frac{5}{2}$

(d) $\frac{9}{2}$

(e) $\frac{13}{2}$

3. The series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+4}}$

(a) diverges by the Alternating Series Test

(b) diverges

(c) converges by the Integral Test

(d) converges conditionally

(e) converges absolutely

4. Evaluate

$$\sum_{n=1}^{\infty} 5\left(\frac{2}{3}\right)^n + \sum_{n=1}^{\infty} \frac{4}{n(n+2)}.$$

(a) 13

(b) 18

(c) 14

(d) 16

(e) 15

5. The interval of convergence for the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-4)^n}{n\,9^n}$ is

(a) $(-5, 13)$

(b) $(4, 9]$

(c) $(-5, 13]$

(d) $[4, 9]$

(e) $[-5, 13]$

6. Evaluate

$$\int_0^{\sqrt{3}/2} \frac{t^2}{(1-t^2)^{3/2}}\, dt.$$

(a) $\sqrt{3} + \dfrac{\pi}{6}$

(b) $\sqrt{2} + \dfrac{\pi}{6}$

(c) $\sqrt{3} - \pi$

(d) $\sqrt{3} - \dfrac{\pi}{3}$

(e) $\sqrt{2} + \dfrac{\pi}{3}$

7. The arc length of the graph of the function $f(x) = \dfrac{1}{2}\left(e^x + e^{-x}\right)$ over the interval $[0, 2]$ is

(a) $\dfrac{1}{2}\left(e^2 - \dfrac{1}{e^2}\right)$

(b) $\dfrac{1}{2}\left(e - \dfrac{1}{e}\right)$

(c) $\dfrac{1}{4}\left(e^2 + \dfrac{1}{e^2}\right)$

(d) $\dfrac{1}{4}\left(e + \dfrac{1}{e^2}\right)$

(e) $\left(e^2 + \dfrac{1}{e}\right)$

8. Using a second degree Maclaurin polynomial of the function $f(x) = \sec x$, the approximation of the value $\sec 1$ is equal to

(a) $\frac{7}{2}$

(b) $\frac{9}{2}$

(c) $\frac{5}{2}$

(d) $\frac{11}{2}$

(e) $\frac{3}{2}$

9. The volume of the solid generated by revolving the region bounded by the graph of $y = 1 - x^2$ (in the first quadrant) about the x-axis is

(a) $\frac{8\pi}{15}$

(b) $\frac{4\pi}{5}$

(c) $\frac{4\pi}{15}$

(d) $\frac{2\pi}{15}$

(e) $\frac{8\pi}{5}$

10. Which one of the following series is/are divergent?

$$\text{(I)} \sum_{n=1}^{\infty} \frac{9^n}{n^5} \qquad \text{(II)} \sum_{n=1}^{\infty} \frac{(2n)!}{n^5} \qquad \text{(III)} \sum_{n=1}^{\infty} \left(\frac{3n+2}{n+3}\right)^n$$

(a) II and III only

(b) I and II only

(c) I and III only

(d) I, II and III

(e) III only

11. Consider the improper integral

$$\int_4^\infty \frac{dx}{x(\ln x)^3}.$$

It

(a) converges to $\dfrac{1}{2(\ln 2)^2}$

(b) converges to $\dfrac{2}{(\ln 4)^2}$

(c) converges to $\dfrac{1}{(\ln 4)^2}$

(d) converges to $\dfrac{1}{2(\ln 4)^2}$

(e) diverges

12. Which one of the following series is/are divergent?

$$\text{(I)} \sum_{n=1}^{\infty} \frac{2+\cos n}{\sqrt{n}} \qquad \text{(II)} \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2+1}} \qquad \text{(III)} \sum_{n=1}^{\infty} \frac{n^k-1}{n^k+1},\ k>2$$

(a) II only

(b) I and III only

(c) I only

(d) II and III only

(e) I and II only

13. Which one of the following series is/are convergent?

$$\text{(I)} \sum_{n=1}^{\infty} \frac{\arctan n}{n^2+1} \qquad \text{(II)} \sum_{n=1}^{\infty} 3^{-n} \qquad \text{(III)} \sum_{n=1}^{\infty} \frac{1}{n^{1.03}}$$

(a) I, II and III

(b) I and III only

(c) I and II only

(d) III only

(e) II and III only

14. The sequence whose nth term is $a_n = \dfrac{\ln(n^3)}{2n}$ converges to

(a) $\dfrac{2}{3}$

(b) $\dfrac{3}{2}$

(c) 3

(d) 2

(e) 0

15. Evaluate

$$\int \frac{x^2 + 12x + 12}{x^3 - 4x}\, dx.$$

(a) $\ln|x-2| - \ln|x+2| - \ln|x| + C$

(b) $5\ln|x-2| - \ln|x+2| + \ln|x| + C$

(c) $\ln|x-2| + \ln|x+2| - 3\ln|x| + C$

(d) $5\ln|x-2| + \ln|x+2| - 3\ln|x| + C$

(e) $5\ln|x-2| - \ln|x+2| - 3\ln|x| + C$

16. A power series representation for the function $f(x) = \ln(x^2+1)$ centered at $c = 0$ is

(a) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n+1},\ [-1,1]$

(b) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1},\ [-1,1]$

(c) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+2}}{n},\ [-1,1]$

(d) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n+1},\ [-1,1]$

(e) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{n+1},\ [-1,1]$

17. Consider the improper integral

$$\int_0^8 \frac{3}{\sqrt{8-x}}\, dx.$$

It

(a) converges to $\sqrt{2}$

(b) converges to $12\sqrt{2}$

(c) diverges

(d) converges to $4\sqrt{2}$

(e) converges to $6\sqrt{2}$

18. The least number of terms required to approximate the sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ with an error of less than 0.01 is (Hint: use the Remainder Theorem.)

(a) 6

(b) 2

(c) 10

(d) 8

(e) 4

19. Evaluate

$$\int \frac{\ln x}{x^3}\, dx.$$

(a) $\frac{1}{x^2}\ln x - \frac{1}{4x^2} + C$

(b) $\frac{-1}{2x^2}\ln x + \frac{1}{x^2} + C$

(c) $\frac{-1}{2x^2}\ln x - \frac{1}{4x^2} + C$

(d) $\frac{-1}{x^2}\ln x + \frac{1}{2x^2} + C$

(e) $\frac{-1}{x^2}\ln x + \frac{1}{x^2} + C$

20. The coefficient of x^{11} in the Maclaurin series for the function $f(x) = x\sin(x^2)$ is

(a) $\frac{-1}{5!}$

(b) $\frac{1}{4!}$

(c) $\frac{1}{5!}$

(d) $\frac{-1}{4!}$

(e) $\frac{1}{9!}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$u = e^{2x} + e^{-2x}$: $\frac{1}{2}\ln u\big\|_2^{e^2+e^{-2}} = \frac{1}{2}\ln\frac{e^2+e^{-2}}{2}$.
2	**(d)**	Intersections $x = -2, 1$; $A = \int_{-2}^{1}(2 - x - x^2)dx = \frac{9}{2}$.
3	**(d)**	Alternating series test passes; $\sum \frac{1}{\sqrt{n+4}}$ diverges ($p = \frac{1}{2}$): conditional.
4	**(a)**	Geometric sum = 10; telescoping sum = $2(1 + \frac{1}{2}) = 3$; total 13.
5	**(c)**	$R = 9$ about 4; $x = 13$ converges, $x = -5$ diverges: $(-5, 13]$.
6	**(d)**	$t = \sin\theta$: $\int_0^{\pi/3}\tan^2\theta\,d\theta = \sqrt{3} - \frac{\pi}{3}$.
7	**(a)**	$f = \cosh x$, $L = \int_0^2 \cosh x\,dx = \sinh 2 = \frac{1}{2}(e^2 - e^{-2})$.
8	**(e)**	$P_2(x) = 1 + \frac{x^2}{2}$, so $\sec 1 \approx \frac{3}{2}$.
9	**(a)**	$V = \pi\int_0^1(1 - x^2)^2dx = \frac{8\pi}{15}$.
10	**(d)**	Ratio: (I) $\to 9$, (II) $\to \infty$; root: (III) $\to 3$; all diverge.
11	**(d)**	$u = \ln x$: $\left[-\frac{1}{2u^2}\right]_{\ln 4}^{\infty} = \frac{1}{2(\ln 4)^2}$.
12	**(b)**	(I) $\geq \sum n^{-1/2}$ diverges; (II) $< \sum n^{-2}$ converges; (III) terms $\to 1$ diverges.
13	**(a)**	(I) compare with $\sum \frac{1}{n^2}$; (II) geometric $r = \frac{1}{3}$; (III) $p = 1.03 > 1$: all converge.
14	**(e)**	$a_n = \frac{3\ln n}{2n} \to 0$ (L'Hopital).
15	**(e)**	Partial fractions $-\frac{3}{x} + \frac{5}{x-2} - \frac{1}{x+2}$, so $5\ln\|x-2\| - \ln\|x+2\| - 3\ln\|x\| + C$.
16	**(e)**	$f' = \frac{2x}{1+x^2}$, integrate the geometric series: $\sum \frac{(-1)^n x^{2n+2}}{n+1}$.
17	**(b)**	Antiderivative $-6\sqrt{8-x}$, limit gives $6\sqrt{8} = 12\sqrt{2}$.
18	**(e)**	$\|R_N\| \leq \frac{1}{(N+1)!} < 0.01 \Rightarrow (N+1)! > 100 \Rightarrow N = 4$.
19	**(c)**	Parts with $u = \ln x$: $-\frac{\ln x}{2x^2} - \frac{1}{4x^2} + C$.
20	**(c)**	$x\sin x^2 = \sum \frac{(-1)^k x^{4k+3}}{(2k+1)!}$; $k = 2$ gives $\frac{1}{5!}$.